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Data Science Capstone - Annotated Bibliography
Professor Hardin
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This project will draw on two complementary strands of existing work. The first is empirical research on higher-education staffing and cost patterns, which I will use primarily for methodology. The second is Ginsberg's institutional account of administrative growth in American universities, which I will use to motivate the hypotheses I test empirically rather than as a data source itself. My primary data will come from IPEDS and ProPublica's Nonprofit Explorer (IRS Form 990 filings), supplemented by the AAUP Faculty Compensation Survey and BEA Regional Price Parities for the cost-of-living adjustment.

Sources

Desrochers, D. M., & Kirshstein, R. (2014). *Labor intensive or labor expensive? Changing staffing and compensation patterns in higher education*. Delta Cost Project, American Institutes for Research.

<https://www.air.org/sites/default/files/downloads/report/DeltaCostAIR-Labor-Expensive-Higher-Education-Staffing-Brief-Feb2014.pdf>

This issue briefly documents a long-run national shift in higher-ed staffing toward non-instructional professional and administrative positions, using IPEDS data from the 1990s through the 2010s. I will use it as the closest methodological precedent for building comparable staffing/compensation categories from IPEDS over time, and its national findings give me a benchmark to test whether my 19-college panel of elite liberal arts institutions follows or departs from the broader pattern.

Hedrick, D., Wassell, C., & Henson, S. (2009). Administrative costs in higher education: How fast are they really growing? *Education Economics*, 17(1), 123–137.

A peer-reviewed econometric study testing whether administrative cost growth in higher education outpaces instructional cost growth, using panel regression methods on IPEDS finance data. I will model my own panel/fixed-effects specification on theirs, extending it with a cost-of-living adjustment and a rank-based breakdown of faculty that their institution-level finance data doesn't include.

Ginsberg, B. (2011). *The fall of the faculty: The rise of the all-administrative university and why it matters*. Oxford University Press.

Ginsberg argues that administrative growth in American universities reflects a structural shift in institutional priorities that has come partly at the expense of faculty. I will use this as the conceptual framing for the project's motivation and as a source of testable hypotheses, while treating its claims as something my empirical analysis should test rather than assume.

Data Sources

List of 19 colleges (used by the Pomona board, provided by Professor Steinberger):

Bowdoin, Bryn Mawr, Carleton, Claremont McKenna, Davidson, Grinnell, Harvey Mudd, Haverford, Middlebury, Oberlin, Occidental, Pitzer, Pomona, Scripps, Smith, Swarthmore, Wellesley, Wesleyan, Williams

Integrated Postsecondary Education Data System (IPEDS), National Center for Education Statistics. <https://nces.ed.gov/ipeds/>

Primary source for historical faculty salary by academic rank, aggregate administrative/managerial compensation, staff headcounts by occupational category (including by tenure status), enrollment, and institutional characteristics for all 19 colleges. Note: IPEDS reports faculty salary by rank, not tenure status directly, so I will use rank (e.g., Assistant Professor) as a proxy for pre-tenure status, cross-checked against IPEDS's separate tenure-status headcount tables; administrator compensation is available only as a single institution-wide aggregate, not by named position. I will build a scraper to assemble a historical panel, since IPEDS's interface only exposes one survey year at a time.

ProPublica Nonprofit Explorer (IRS Form 990 filings). <https://projects.propublica.org/nonprofits/>

All 19 colleges are private nonprofits and file Form 990s, whose Part VII reports compensation for officers, trustees, key employees (>\$150k), and the five highest-compensated other employees (>\$100k) by name and title. I will use this to track named senior-administrator positions (e.g., president, provost, CFO) over time, while recognizing it captures only top-of-house pay, not the broader administrative/professional staff population. Historical coverage is strongest from roughly the 2010s onward, since older filings predate mandatory e-filing and exist mainly as scanned images rather than structured data.

American Association of University Professors (AAUP) Faculty Compensation Survey / Annual Report on the Economic Status of the Profession.

<https://www.aaup.org/issues-higher-education/faculty-employment/faculty-compensation-survey>

A longer-running annual survey of faculty salary by rank, used as a cross-check on IPEDS faculty figures and to extend the panel further back than IPEDS microdata reliably allows. Institution participation is voluntary, so I will confirm coverage for each of the 19 colleges before relying on it as a primary series.

U.S. Bureau of Economic Analysis, Regional Price Parities by State and Metro Area.

<https://www.bea.gov/data/prices-inflation/regional-price-parities-state-and-metro-area>

Provides the regional cost-of-living index used to convert nominal salaries into cost-of-living-adjusted (real) compensation. This series begins in 2008, so regional cost-of-living adjustment will be applied from 2008 onward; earlier years will be inflation-adjusted using national BLS CPI-U rather than a regional index.